

MODULE 10

Cybersecurity Principles, Practices, and Processes

Endpoint Security | Cisco Networking Academy

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Cybersecurity Essentials

Topics Covered:

- 10.0 Introduction to Cybersecurity Principles
- 10.1 The Three Dimensions — The Cybersecurity Cube
 - 10.1.3 CIA Triad — The Principle of Confidentiality
 - 10.1.4 Protecting Data Privacy
 - 10.1.6 Data Integrity | 10.1.7–10.1.9 Availability

10.0 Introduction

Foundations of Cybersecurity Principles

10.0

Why Cybersecurity Principles Matter

Organizations must protect data, systems, and users from an ever-growing threat landscape. Understanding core cybersecurity principles forms the foundation for building effective defenses.

CONFIDENTIALITY

Ensuring information is accessible only to those authorized to access it.

INTEGRITY

Safeguarding accuracy and completeness of data and processing methods.

AVAILABILITY

Ensuring authorized users have access to information and systems when needed.

These three pillars form the CIA Triad — the cornerstone of all cybersecurity frameworks and standards worldwide.

10.1 The Three Dimensions

Understanding the Cybersecurity Cube

10.1

The Cybersecurity Cube

John McCumber developed the Cybersecurity Cube as a model to help identify the three dimensions that must be addressed to achieve security.

DIMENSION 1 Security Goals

Confidentiality · Integrity · Availability (CIA Triad). These are the core security goals that every organization strives to protect.

DIMENSION 2 Data States

Data in Process · Data in Transit · Data at Rest. Security controls must address data in all three states.

DIMENSION 3 Safeguards / Countermeasures

Technology · Policy & Practices · People & Education. Layered defenses combining tools, policies, and human training.

10.1.2 The Cybersecurity Cube

A Three-Dimensional Security Model

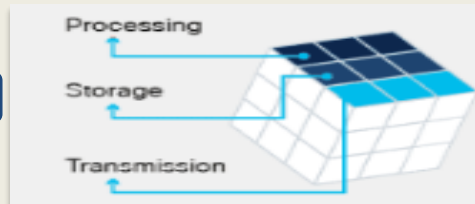
10.1.2

McCumber Cube — Three Dimensions

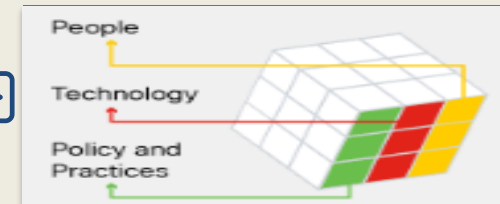
Security Goals



Data States



Safeguards



Goal: Confidentiality

Prevent unauthorized disclosure of information through encryption, access controls, and data classification.

State: Data in Process

Data actively being used in computations, applications, or transactions — must be protected in memory.

Goal: Integrity

Prevent unauthorized modification using hashing, digital signatures, and change-control procedures.

State: Data in Transit

Data moving across networks — protect with TLS/SSL, VPNs, and encrypted communication channels.

Goal: Availability

Ensure reliable access via redundancy, failover systems, and disaster recovery planning.

State: Data at Rest

Stored data on disks, databases, or removable media — protect with full-disk encryption and access controls.

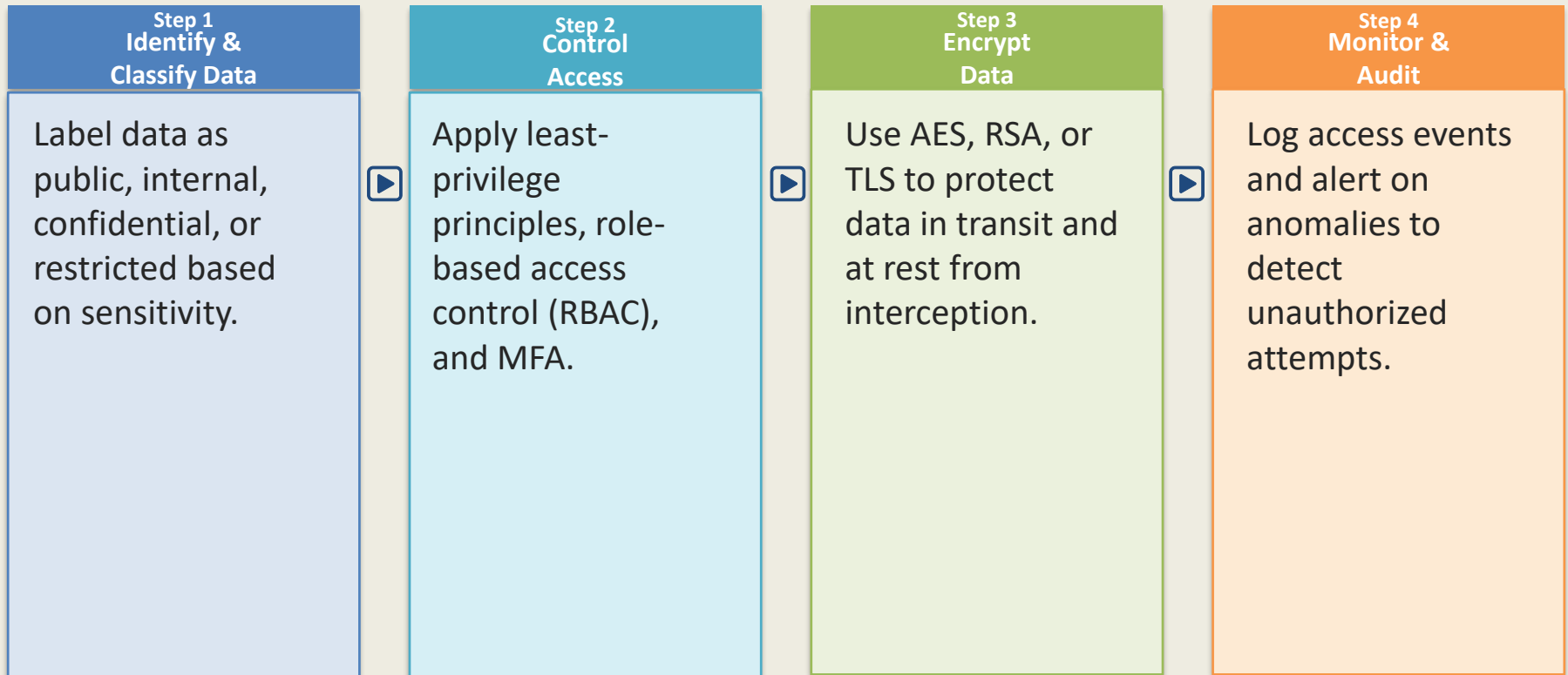
10.1.3 CIA Triad — Confidentiality

The Principle of Confidentiality

10.1.3

Protecting Information from Unauthorized Access

Confidentiality ensures that sensitive information is accessed only by authorized individuals. Various techniques are applied at different layers.



10.1.4 Protecting Data Privacy

Privacy Principles and Techniques

10.1.4

Data Privacy vs. Data Security

Privacy focuses on appropriate data use and user rights, while security provides the technical controls that enforce those rights.

Data Minimization

Collect only the data that is strictly necessary for the stated purpose. Avoid storing sensitive information longer than needed.

Anonymization & Pseudonymization

Remove or mask identifying information so individuals cannot be identified from the dataset without additional key data.

Consent & Transparency

Inform users about data collection practices. Obtain explicit consent before processing personal data. Comply with GDPR and similar laws.

Data Access Controls

Restrict who can view, edit, or share personal data. Apply role-based access, audit trails, and data-loss prevention (DLP) tools.

10.1.6 Data Integrity

Ensuring Accuracy and Trustworthiness of Data

10.1.6

What Is Data Integrity?

Integrity guarantees that data has not been altered in an unauthorized manner — whether in storage, during processing, or in transit.

Hashing

MD5, SHA-256, SHA-3 — generate a fixed-size digest; any alteration changes the hash, revealing tampering.

Digital Signatures

Combine asymmetric encryption with hashing to verify both the source and the integrity of a message or document.

Checksums & CRCs

Lightweight error-detection codes appended to data blocks to catch accidental corruption during transmission or storage.

Change-Control Procedures

Version control, audit logs, and approval workflows ensure only authorized modifications are made to data and systems.

Database Integrity Constraints

Primary keys, foreign keys, NOT NULL, and UNIQUE constraints enforce relational integrity at the database level.

10.1.7 & 10.1.9 Availability

Ensuring Systems Are Accessible When Needed

10.1.7

Ensuring Availability

Availability means authorized users can access systems and data reliably. Threats include DoS attacks, hardware failure, and natural disasters.

Redundancy

Duplicate hardware, power supplies, and network links eliminate single points of failure.

Backups & Recovery

Regular automated backups with tested restore procedures to minimize downtime.

Failover Systems

Hot/warm/cold standby servers automatically take over when primary systems fail.

Load Balancing

Distribute traffic across multiple servers to prevent overload and maintain performance.

DoS / DDoS Mitigation

Rate limiting, traffic scrubbing, and CDN services absorb volumetric attacks.

Disaster Recovery Plan

Documented RTO/RPO targets with tested procedures to restore operations quickly.

Module 10 — Summary

10.0 Intro	Cybersecurity principles protect data, systems, and users across all industries.	10.1.3 Confidentiality	Data classification, access control, encryption, and monitoring prevent unauthorized disclosure.
Cube Dimension 1	Security Goals: Confidentiality, Integrity, and Availability form the CIA Triad.	10.1.4 Privacy	Minimization, anonymization, consent, and DLP controls protect personal data and ensure compliance.
Cube Dimension 2	Data States: In Process, In Transit, and At Rest each require tailored protections.	10.1.6 Integrity	Hashing, digital signatures, and change-control procedures ensure data accuracy and trustworthiness.
Cube Dimension 3	Safeguards: Technology, Policy, and People/Education work together as layered defenses.	10.1.7/9 Availability	Redundancy, backups, failover, load balancing, and DDoS mitigation keep systems online and accessible.